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Chapter 1

Getting Started



Prerequisites

Please note that System 21 Constructor supports only the configuration detailed separately on the latest System 21 Configuration Information Sheet issued by JBA Software Products Limited, although individual configurations can be agreed by/with clients and JBA Branches for local support of the product.

It is assumed that you are familiar with the standard elements of the Microsoft Windows interface, such as the controls for opening and closing windows, scroll bars, buttons and icons. However, help is included for some of the less well known Windows terms and techniques.



Tip: For more information about Windows you should consult the Windows documentation.

Opening And Closing Constructor

Opening

Please open your Start Menu and select JBA System 21 Constructor from your program list. Constructor immediately displays the Workspace for you to start viewing or designing Business Processes and Activities.



Note: You do not log on to a System 21 host machine unless you execute one of the components in your business system. You do this either by double clicking on individual symbols on the Workspace or by using the Execute button on the toolbar. Please refer to the System Logon Window.



Online Help is available within the software itself as an option on the main menu.

Closing

Before closing Constructor please exit all activities or applications, and end any host sessions: no window should be open or iconised on the Workspace. If a process is still running at the server this may lead to data loss.



Note: an application or activity has been exited when the window that contains it is closed.

To close Constructor use the normal Windows functions to close the program, for example select Exit from the File Menu on the main menu bar.

If you have made any changes to a Design then Constructor will prompt you to save those changes.

What Is Constructor?

Definition

Constructor is a framework generator tool that enables you to construct your business system using a hierarchy of components.

Using a Windows environment you model your system design on a Workspace, incorporating System 21 application components. You decide the components and then link them together to form a logical business structure.

Components are grouped hierarchically:

- The top level in the hierarchy consists of the Business Processes.
- The middle level consists of the Assembled Activities, Elemental Activities, Activity Lists and Subassemblies.
- The bottom level consists of the Business Methods.

A design should ideally be started at the bottom level so that the top level finally incorporates all the lower levels.

Please refer to the Components Hierarchy Diagram.

Constructor integrates fully with Enterprise Work Management and System 21 Explorer.



Note: You do not have to understand or implement Enterprise Work Management in order to use Constructor.

Objectives

You can use Constructor in one of three ways:

- As a design tool for the construction of Assembled Activities.
- As a design tool for Enterprise Modelling.
- As a graphical design/configuration tool for Enterprise Work Management. You configure the design of a Business Process and Activate it for the Enterprise Work Management Engine. Enterprise Work Management then controls the flow of work through Explorer by initiating activities and managing Action Lists.

System 21 enables a customer to integrate standard System 21 components in conjunction with their own bespoke components, based on Microsoft ActiveX technologies.



Tip: This guides explains the Menu options, Toolbar buttons and Windows relating to Enterprise Work Management functionality. However, for detailed tutorials which explain how to build Business Processes and how to enable existing activities, please refer to the System 21 Enterprise Work Management Guide.

What You Can Do

At each stage in the design process you can validate the design so that you be certain that it is viable.

You can preview and execute Business Methods, Activities, Activity Lists and Subassemblies from within Constructor.

Although you cannot directly execute a Business Process in Constructor you can perform a walkthrough to verify the flow of control between the configured components. You can launch the underlying System 21 application components individually.

What To Do First - Tutorial

Constructor is a comprehensive design tool, however, you can use it at different levels according to your needs and experience.

This tutorial takes you step-by-step through the basic functions to get you started. This guide also gives you help for every menu option and every field on each screen: you just select the Help option from the Main menu.

When you feel ready you can go on to the Enterprise Work Management guide and work through the more advanced tutorials and learn how to recreate the Active Enterprise Business Processes and enable existing Elemental Activities for Work Management.

Please follow these steps:

Step 1 - What You See First

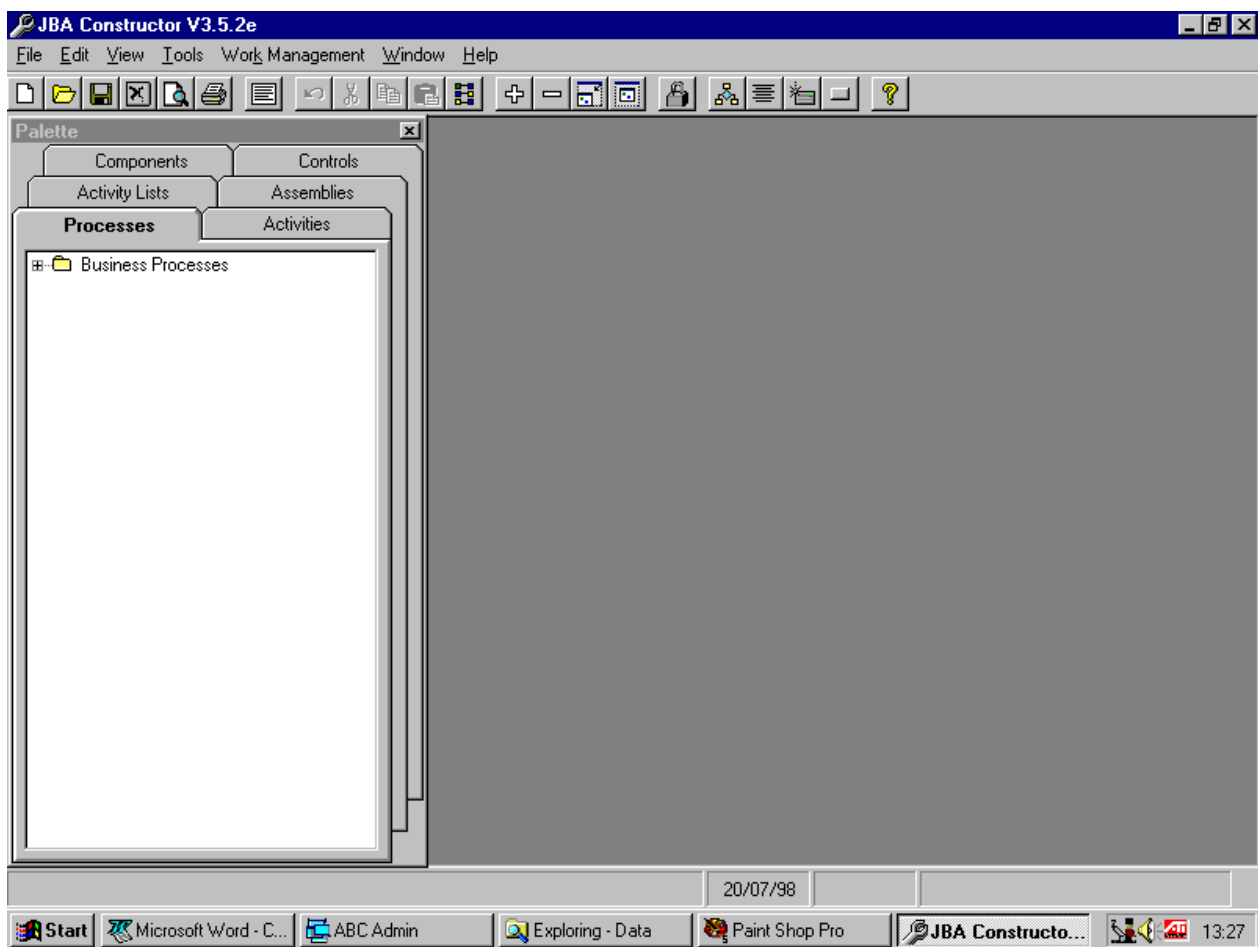
When you open Constructor you straightaway see many of the screen elements that you need to construct your business system. You see these elements, in this order, starting from the top of the screen: Title Bar; Main Menu Bar; Toolbar; Display Area; Status Bar.

The only screen elements that you cannot see are context sensitive pop up menus, Constructor windows, application windows and error messages. You will use these when you start designing your business system.



Tip: Please take a minute now to familiarise yourself with these by reading the Help text subsections below.

The Whole Screen



The Screen Parts

Starting from the top of the screen:

Title Bar

Tells you at first which version of Constructor you are using. For example, JBA Constructor V3.5.2e. Later it will tell you the name of your design if the current window is maximised.

Main Menu Bar

File Edit View Tools Work Management Window Help

Six options, each with their own separate Help section.

Toolbar



If you have not seen Constructor before then some of the buttons on this toolbar will be new to you. There is a separate Help section where each button on the Toolbar is explained in detail.



Note: When you design a Business Process the Preview and Execute buttons are replaced by the Walkthrough button. 

Display Area

The Constructor display area is the area below the toolbar.

It is divided into two parts:

Component Browser Area on the left

This area contains tabbed folders of components. This is where you are going to select all the components that you need for your business system.

Workspace Area on the right

This is where you Review or Design your business system. This is where the actual design sits on the screen and where you can work with it.

Status Bar

Process1	23/03/98	Business Process
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The Status Bar is the area at the bottom of the screen, and is split into three sections:

- Name of component
- Date
- Type of component, for example, Business Process, Activity or Activity List.

Step 2 - Understanding The Components

In order to construct your business system you use components from a hierarchy of components and link them together to form a logical business structure.

You select the components from the Component Browser Area and put them on the Workspace. Each component has a symbol for easy recognition. Sometimes the component has two symbols: one on the Component Browser Area and one on the Workspace.

The component symbols and their hierarchy form the basis of Constructor's design capabilities.

Please refer to the Components Hierarchy Diagram.



Tip: It is important to understand these. However, we recommend that you skim through the Help text subsections below, about The Symbols and The Hierarchy, get a general understanding now but then use them for reference later.

The Symbols



Tip: You might want to print this Help page for reference later.

You see these symbols on both the Component Browser Area and the Workspace unless otherwise stated:

Controls

- | | |
|--|--|
|  User Selected Path | |
|  Parallel Path | |
|  Start Condition | |
|  End Condition | |
|  Manual Activity in the Browser |  Manual Activity on the Workspace |
|  PC Short Cut in the Browser |  PC Short Cut on the Workspace |
|  Action Agent in the Browser |  Action Agent on the Workspace |

└─ Connections

Business Processes

-  Business Process in the Component Browser Area or Subassembly in the Component Browser Area
-  Business Process on the Workspace
-  Business Process on the Workspace where there is more than one Start Condition. Please refer to Nesting Business Processes

Activities

-  Assembled Activity in the Component Browser Area
-  Assembled Activity on the Workspace or Subassembly on the Workspace
-  Elemental Activity

Business Methods

-  Business Method

Note that more nodes are added when on the Workspace

 Activity Lists on the Component Browser Area only

Note that the Business Processes, Assembled Activities and Subassemblies all have a cross in the centre of the symbol, when on the Workspace, to indicate more subprocesses. You can double click on the cross and view the subcomponents.

All the symbols on the Workspace, except Connection, have circular segments to the right side of the centre circle. These are referred to as Nodes. These nodes represent the exit points from the component, for example for Save and Cancel buttons.

To amend the node details use the Properties Window.



Warning: When you amend Properties details you update the component details on the design-time database, not just the details for this design. This would be reflected in all future designs for all users of that component.



Note: Activity Lists contain activities which have only one node each. This is because Activity Lists are a series of Elemental Activities to be processed sequentially. They each have only one exit node so that as soon as one activity in the list is completed the processing moves straight onto the next activity.

The Hierarchy



Tip: You might want to print this Help page for reference later.

Please refer to the Components Hierarchy Diagram.

The components are organised in a strictly hierarchical manner. There are three levels in the basic hierarchy:

Business Processes

These are at the top level of the business function and are broken down into Elemental Activities, Assembled Activities and other Business Processes.

Please refer to Nesting Business Processes.

Activities



Note: If you are using Enterprise Work Management please note that it is the Elemental Activities which you enable. These represent RPG Programs.

There are two types of Activities:

- Assembled Activities
- Elemental Activities.

Business Methods

These are the lowest level components, representing subactivities and cannot be broken down any further.



Note: The Business Processes and Assembled Activities have a cross in the centre of the symbol to indicate that further function lies behind. To view that function either double click on the cross or select Preview from the pop up menu.

There are two other types of components that are not strictly within the basic hierarchy:

Activity Lists

These are a series of Elemental Activities to be processed sequentially. They each have only one exit node so that as soon as one activity in the list is completed the processing moves straight onto the next activity.

Subassemblies

A logical collection of Business Methods. They are used to group together the more frequently used Business Methods to save time when designing. You find Subassemblies within Assembled Activities.

There are two components which are conceptual and used for the design:

Process Controls

These are used to mark the start and end of the Business Processes and Activities and to influence the pathways between components.

The Connections

These are created in design mode and link the components together to form a continuous process.

Step 3 - Using The Component Browser Area



Tip: Please read through this and try the suggestions below.

The Component Browser Area contains tabbed folders of components. This is where you are going to select all the components that you need for your business system. The components are organised within the Browser in the same way that files and folders are organised within your Microsoft Windows Explorer.



Note: JBA supply you with a set of initial components but you can of course create your own.

The structure that you see in the Component Browser Area, reflects the contents of the design-time database. Mostly this consists of Elemental Activities and as they represent RPG programs you cannot view these under Program Files, with Windows Explorer.



Note: You can move activities from one folder to another in the Component Browser Area or create new folders of your own, using a context sensitive right click menu. It is not recommended that you do this at this stage.

You can create Assembled Activities, Activity Lists and Subassemblies and these initially appear directly under the Activities folder. You can rearrange them as you wish. You can also view these under Program Files\JBASys21, with Windows Explorer.

Tabs

The tabs on the folders are arranged at first like this:



However, as you select each tab the order can change.

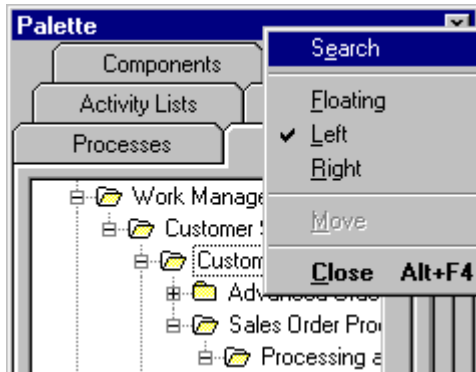
Searching



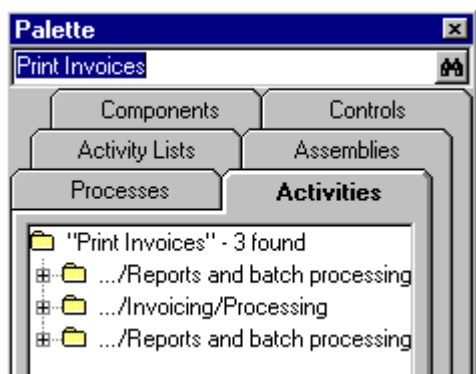
Tip: This is a useful facility so please try this now.

You can search for the name of any component in the Component Browser Area.

1. Move the cursor over the title bar in the Component Browser Area.
2. Right click and select Search from the pop up menu.



3. Enter the name for the search in the space provided. This could be the textual name or the shortcut code.
4. Press return and the search results are displayed in place of the Component Browser.



5. To return to the Component Browser Area move the cursor over the title bar again right click and deselect Search.

Tips

Note that the way in which you put components onto the Workspace varies according to the type of component.

Open each of the tabs in turn, noticing the organisation of groups of components. At the same time read each of the subsections of the Help below. There is a subsection for each tab.

Experiment with the existing components. This Help section tells you how to open them up into new windows on the Workspace and find out which ones you can insert into others and which ones can be nested.

Processes Folder

This contains groups of Business Processes.

There are four ways to open a Business Process and create a new window on the Workspace:

- Drag the symbol from the Component Browser Area to a blank part of the Workspace.
- Double click the symbol in the Component Browser Area.
- Select Open from the File Menu and then select Processes from the Open Entity Window.
- Select the Open button from the Toolbar.

Please refer to Nesting Business Processes.

Activities Folder

This contains groups of both Elemental Activities and Assembled Activities.

Elemental Activities

The Elemental Activities represent RPG Programs and are organised into groups. You could have these:

- Applications
- SCOR Inputs and Outputs
- Enterprise Work Management Enabled.

Elemental Activities cannot be broken down any further and the only way to put them on the Workspace is to drag the symbol from the Component Browser Area to an existing design on the Workspace.

You cannot nest Elemental Activities.



Note: Please refer to the Enterprise Work Management Guide for tutorials explaining how to enable existing Elemental Activities.

Assembled Activities

The Assembled Activities are displayed under the top level grouping called Activities. These can contain Elemental Activities, Business Methods and Subassemblies.

You can open an Assembled Activity and create a new window on the Workspace. There are four ways in which to do this:

- Drag the symbol from the Component Browser Area to a blank area on the Workspace.
- Double click the symbol in the Component Browser Area.
- Select Open from the File Menu and then select Activities from the Open Entity Window.
- Select the Open button from the Toolbar.

You cannot nest Assembled Activities.

You can insert an Elemental Activity into an Assembled Activity.

You can insert an Assembled Activity into a Business Process.

Lists Folder

This contains groups of Activity Lists.

You can open an Activity List and create a new window on the Workspace. There are four ways in which to do this:

- Drag the symbol from the Component Browser Area to a blank area on the Workspace.
- Double click the symbol in the Component Browser Area.
- Select Open from the File Menu and then select Lists from the Open Entity Window.
- Select the Open button from the Toolbar.

You cannot nest Activity Lists or insert them into designs for other components.

Assemblies Folder

This contains groups of Subassemblies.

You can open a Subassembly and create a new window on the Workspace. There are four ways in which to do this:

- Drag the symbol from the Component Browser Area to a blank area on the Workspace.
- Double click the symbol in the Component Browser Area.
- Select Open from the File Menu and then select Assemblies from the Open Entity Window.
- Select the Open button from the Toolbar.

You cannot nest Subassemblies but you can insert them into designs for Assembled Activities.

Components Folder

This contains groups of Business Methods.

Business Methods cannot be broken down any further and the only way to put them on the Workspace is to drag the symbol from the Component Browser Area to an existing design on the Workspace.

You can insert them into Assembled Activities or Subassemblies.

You cannot nest Business Methods.

Controls Folder

This contains the process control tools. They cannot be opened but have to be dragged onto a new or existing design:

- Start Condition - drag this onto a new design at the beginning.
- End Condition - drag this onto a new design to complete the design.
- User Selected Path - drag this onto a design when you want to split the logical flow into separate pathways.
- Parallel Path - drag this onto a design when you insert a piece of parallel processing.
- Manual Activity - drag this onto a design when you want to allow the user to have an input to the process. For example, where the user decides what to do about failed credit checks. You can also insert a new Manual Activity using the Workspace Menu.
- PC Short Cut - drag this onto a design where you want to enable the user to insert a piece of programming function into the design.
- Action Agent - drag this onto a design when you want to insert a hidden piece of programming. For example, to tell the Business Process how to handle credit checking after the processing for Order Entry. You can also insert a new Action Agent using the Workspace Menu.

Step 4 - Designing



Tip: Please experiment with the suggestions and steps in the subsections of Help text below to get you going. It is a good idea to work with existing designs first before creating a new one.

Work With An Assembled Activity



Tip: Start by opening an Assembled Activity.



Note: You cannot Activate an Activity for Enterprise Work Management.

Note the things you can do with an Assembled Activity and try them now:

Design It

You could design the activity but this has been done for you.

Validate It

Select the Validate button on the Toolbar.  This lists any errors in the logical design.

Preview Or Execute It

Select the Preview or Execute button on the Toolbar.   Both these take you logically through the whole process stopping at each Elemental Activity, Business Method or control symbol.

However, Execute actually executes the components. It depends what the component does: it may trigger a server based process or one that is purely local to the PC.

Each time the Preview or Execution stops you get a window of details for that component:

- At the beginning you get the Start Details Window.
- Whenever you hit a User Selected Path control symbol you get the User Selected Path Window.
- Whenever you hit an Elemental Activity or Business Method the component is executed, provided that it is detected. If the Elemental Activity or Business Method does not exist you get the Properties Window.
- At the end you get the End Details Window.

You can amend details for the component or continue.



Warning: When you amend Properties details you update the component details on the design-time database, not just the details for this design. This would be reflected in all future designs for all users of that component.

Work With A Business Process



Tip: Start by opening a Business Process.

Note the things you can do with Business Processes and try them now.



Note: There are restrictions on the way in which you construct Business Processes depending on whether you use Constructor as a design tool for enterprise modelling or a configuration tool for Enterprise Work Management. Activating for Enterprise Work Management involves rigorous validation checking.

Design It

You could design the Business Process but this has been done for you. You design what the Enterprise Work Management process is going to be.

Validate It

Select the Validate button on the Toolbar.



This lists any errors in the logical design.

Walkthrough It



Note: Although you cannot directly execute a Business Process in Constructor you can walkthrough it to verify the flow of control between the configured components. You can launch the underlying System 21 application components individually.

Select the Walkthrough button on the Toolbar. 

Walkthrough takes you logically through the whole process stopping at each element of the design. Each time it hits an Assembled Activity or another Business Process it breaks it down into its lowest level components, including all its control icons. If there are Business Processes within Business Processes then all these break down too. For example, you hit a Start Condition control symbol at the beginning of the whole process and then again at the beginning of each subprocess.

Please refer to Nesting Business Processes for further information.

Each time the walkthrough stops you get a window of details:

- Whenever you hit a Start Condition control symbol you get the Start Details Window.
- Whenever you hit a User Selected Path control symbol you get the User Selected Path Window.
- Whenever you hit an Activity symbol you get the Properties Window.
- Whenever you hit an End Condition control symbol you get the End Details Window.

You can amend details for the component or continue.



Warning: When you amend Properties details you update the component details on the design-time database, not just the details for this design. This would be reflected in all future designs for all users of that component.

Activate It For Enterprise Work Management

This is so that Enterprise Work Management can then run it using the Enterprise Work Management Engine. This Activation procedure compiles the configuration for you. To do this you select Activate from the Work Management Menu on the Main Menu Bar. This displays the Activate Window.



Warning: Take care replacing Modes of Execution and other items on the server. You would be overwriting those items created by an activation from another PC. If you have more than one PC user please take advice from your JBA consultant.

Activate does the following things:

- Validate. This performs the same validation as the Validate button. If you previously validated and it failed then you will not be able to Activate either.
- Compiles the configuration.
- Performs ODBC updating of the Enterprise Work Management Engine tables.

Start A New Design

Follow these steps:

Step 1

Select New from the File Menu on the Main Menu Bar, or the new button on the Toolbar.

Step 2



Note: You cannot create a design for an Elemental Activity or Business Method.

Choose one of these from the pop up menu:

- Business Process
- Assembled Activity
- Subassembly
- Activity List.

Step 3

Whichever choice you make you get a window with one field for you to enter the name of the new component.

Enter a name and select the OK button.

Step 4

A new window opens for you with the name at the top. Look in the Windows Menu on the Main Menu Bar and note that your new name is listed there.

Note that the Start Condition and End Condition symbols are placed on the Workspace for you.



Note: This follows Microsoft Windows conventions.

Design A Business Process Or Activity



Tip: Please follow the instructions for Start A New design first of all, selecting a Business Process or Activity.



Warning: As you put components onto the Workspace you can click with the right mouse button on the component and edit details using the pop up menu. However, note that when you amend Properties details you update the component details on the design-time database, not just the details for this design. This would be reflected in all future designs for all users of that component.

Follow these steps:

Step 1

Drag across the various components which you want in your Business Process or activity. Note that you already have a Start Condition and End Condition. It is possible to have multiple Start and End Conditions.

Step 2

Define a sequence for the Business Process or activity by joining the components together with connection lines.

In order to create a connection line you click with the left mouse button on a node of one component, continue to hold the button down, and then release it when the cursor is over the circular part of the target component. Once the components are joined together you have finished the construction of the process.

Step 3

Click on the Validate button on the Toolbar.



This validates the design by checking that the process you have defined meets the rules for the basic design criteria.

Constructor displays the Validation Progress Report Window which displays a list for you to examine and then correct any errors.

The list tells you whether all the output nodes are connected, whether all the input conditions are satisfied, whether all the necessary connections have been made, and whether all the consumables are present and correct.

Step 4

Click on the Save button on the Toolbar to save the Business Process or activity to disk.

Step 5

If you are designing an Activity then click on the Preview button on the Toolbar



and then click on the Execute button on the Toolbar



Note: Execute actually executes components. It depends what the component does: it may trigger a server based process or one that is purely local to the PC.

If you are designing a Business Process then click on the Walkthrough button on the Toolbar.



Warning: If you are designing a Business Process then this has to be executed by the server based workflow engine. You have to Activate the process for Enterprise Work Management. You cannot execute it here.

Step 6



Note: This is for Business Processes only. You cannot Activate an Activity for Enterprise Work Management.

Activate the Business Process for Enterprise Work Management. Select Activate from the Work Management Menu. This displays the Activate Window.



Note: There are restrictions on the way in which you construct Business Processes, depending on whether you use Constructor as a design tool for enterprise modelling or a configuration tool for Enterprise Work Management. Activating for Enterprise Work Management involves rigorous validation checking.

This is so that Enterprise Work Management can then run it using the Enterprise Work Management Engine. This compiles the configuration for you.



Warning: When you amend Properties details you update the component details on the design-time database, not just the details for this design. This would be reflected in all future designs for all users of that component.

Activate does the following things:

- Validate. This performs the same validation as the Validate button. If you previously validated and it failed then you will not be able to Activate either.
- Compiles the configuration.
- Performs ODBC updating of the Enterprise Work Management Engine tables.

Please refer to Nesting Business Processes for further information.

Step 5 - More Constructor Features

These are important features but they are not immediately obvious when you first start to use Constructor.

Context Sensitive Pop Up Menus



Tip: These are an important part of Constructor.
It's worth experimenting to get used to them.

Just as for standard Windows 95 or Windows NT applications, Constructor supplies you with useful context sensitive pop up menus.

To display these you click with the right mouse button on any part of Constructor.

The pop up menus vary depending on which component you choose to click with the right mouse button. Constructor recognises the context from which the menu was requested and goes on to display the appropriate windows for you to enter further details for your configuration.

Constructor Windows

You can display the Constructor windows in several ways:

- By selecting options from the Main Menu Bar
- By selecting options from the context sensitive menus
- By clicking on symbols on the Workspace
- By dragging tools and elements onto the Workspace from the Component Browser Area.

Constructor recognises the context from which the menu was requested and goes on to display the appropriate windows for you to enter further details for your configuration.



Tip: See the separate Help section where all the fields and buttons on each Window are explained in detail.

Application Windows

Each time you execute an activity the system starts a new session. The session number is displayed on the Status Bar. Each new session is displayed as a new window. When the system detects a new window, it centres it on the screen on top of any existing windows.

You can change this by clicking and dragging the windows to separate the overlapping windows. You can use the buttons at the top right hand side of a window to minimise, maximise or close the window.

You can have a maximum of eight windows open, dependent on the system resources available.

Windows Terms And Techniques

It is assumed that you are familiar with the standard elements of the Windows interface such as the controls for opening and closing windows, scroll bars, buttons and icons. However, here is some information about some of the less well known ones.



Tip: For more information about Windows you should consult the Windows documentation.

Spin Button



A Spin Button is a control consisting of two arrows, one pointing upward and one pointing down, usually associated with a field containing numeric characters.

When the upward pointing arrow is selected the associated numeric field is incremented, when the downward arrow is selected the associated numeric field is decremented.

Option Button Group

- ☒ **None**
- ☐ **Description**
- ☐ **Variable Field**

Option button groups represent a group of mutually exclusive options. You can select only one option at a time, and if you already have one option selected your current selection replaces it. The selected option button has a coloured circle in the centre of it.

Option Buttons are sometimes called Radio Buttons.

Check Box



A check box presents non exclusive options - you can select as many check box options as needed. When a check box is selected a cross appears in it and the function it is associated with is enabled.

Drop Down List



A drop down list appears initially as a rectangular box containing the current selection. When you select the down arrow in the square box at the right a list of available choices appears. If there are more items than can fit in the box, scroll bars are provided.

List View

Name	Value	Type
Names 1	10	String
Names 2	20	String
Names 3	30	Integer

A list view is a box containing several lists in columns. You can scroll up and down the list to view it. You cannot edit the details in the list directly. Instead you select a row and the details for that row are displayed in fields above.

Grid Box

A grid box is a JBA listing and viewing tool.

Error Messages

Error messages are displayed in separate windows:



You will not be able to continue input until you clear the error message. In most cases this means clicking the OK button.